AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Design Phase – Solution 2

Project Domain: School / College IT Helpdesk | Platform: ServiceNow

1. System Design

The proposed system is a ServiceNow-based automated ticket classification solution. Its purpose is to identify the type of an IT support request from the incident text and automatically assign suitable Category and Subcategory values. The design focuses on reducing repetitive manual classification and creating a consistent structure for helpdesk incident records.

2. Design Components

Component	Responsibility
Incident Form	Collects Short Description, Description and other incident details.
Flow Designer	Controls the automation sequence and decision branches.
Condition Rules	Checks the incident text for predefined keywords/patterns.
Classification Mapping	Connects issue patterns with Category and Subcategory values.
Incident Record	Stores the final classification and ticket information.

3. Functional Requirements

- The system shall accept an incident created through the ServiceNow Incident process.
- The system shall inspect the Short Description and Description fields.
- The system shall identify common IT issue patterns such as Wi-Fi, projector, password and slow computer problems.
- The system shall assign a predefined Category and Subcategory when a matching condition is found.
- The system shall provide a fallback classification when no rule matches.
- The system shall preserve the original incident information while updating only the intended classification fields.

4. Non-Functional Requirements

- The design should be simple enough for a ServiceNow administrator to maintain.
- The flow should execute consistently for repeated incidents with similar descriptions.
- The solution should be extensible so new issue types can be added later.
- The design should support testing through Flow Designer execution details.

5. System Flow

User Request → Incident Record → Flow Trigger → Text Evaluation → Rule Matching → Category/Subcategory Selection → Incident Update → Classification Complete

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6. Data Flow Design

The data flow begins when an incident is submitted. The incident text becomes the input for the classification logic. Flow Designer evaluates the available information and follows the matching decision branch. Once a classification is selected, the incident record is updated and the automation ends.

Stage	Input / Processing	Output
1. Request	User enters an IT issue.	Incident created.
2. Read Data	Flow reads Short Description and Description.	Text available for evaluation.
3. Match	Conditions compare text with issue keywords.	Matching branch identified.
4. Classify	Mapping rule selects Category/Subcategory.	Classification values selected.
5. Update	Incident record is updated.	Classified ticket.

7. Classification Rule Matrix

Issue	Indicative Keywords	Category	Subcategory
Wi-Fi issue	wifi, wireless, internet, network	Network	Wi-Fi / Connectivity
Projector issue	projector, HDMI, display, projection	Hardware	Projector
Password issue	password, login, reset, credentials	Access	Password / Account
Slow computer	slow, lag, freeze, performance	Hardware	Performance
Unknown issue	no matching pattern	General	Other

8. Flow Designer Implementation Logic

- Trigger: Start the flow when an Incident is created.
- Input: Use the triggering Incident record as the data source.
- Condition 1: Check for network/Wi-Fi related terms.
- Condition 2: If not matched, check for projector/display terms.
- Condition 3: If not matched, check for password/login terms.
- Condition 4: If not matched, check for slow/performance terms.
- Else: Apply the General/Other fallback classification.
- Update: Write the selected Category and Subcategory to the Incident record.

9. Rule Priority Design

Rules should be ordered so that specific conditions are evaluated before broad conditions. This prevents a general keyword from taking precedence over a more meaningful issue classification. The rule order can be modified as the project expands.

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10. User Interaction Design

From the user's perspective, the process remains simple. The user only needs to describe the IT problem in the normal incident fields. Classification is performed by the automation in the background, so the user does not need to manually select every classification value.

11. Sample Input and Output

Sample Incident	System Classification
“Wi-Fi is not working in our classroom.”	Network → Wi-Fi / Connectivity
“Projector shows no display.”	Hardware → Projector
“I cannot login because I forgot my password.”	Access → Password / Account
“Computer is very slow and freezes.”	Hardware → Performance
“I have an issue with another IT service.”	General → Other

12. Test Plan

Test ID	Validation	Expected Result
DP-01	Create Wi-Fi incident.	Correct Network classification.
DP-02	Create projector incident.	Correct Hardware classification.
DP-03	Create password incident.	Correct Access classification.
DP-04	Create slow-computer incident.	Correct Performance classification.
DP-05	Create unmatched incident.	Fallback classification applied.
DP-06	Review Flow execution.	Flow completes without error.

13. Exception & Fallback Design

- Empty or unclear descriptions should follow the fallback branch rather than causing the flow to fail.
- If more than one issue pattern is detected, the configured rule priority should determine the final classification.
- Invalid or unavailable classification values should be handled through controlled flow logic and reviewed during testing.
- Unclassified tickets can be manually reviewed and used to improve future rules.

14. Expected Project Outcome

After implementation, the project should demonstrate that common IT helpdesk incidents can be automatically classified using ServiceNow Flow Designer. The solution provides a repeatable workflow from incident creation to classification, while keeping the classification rules visible and manageable. The design can later be extended with additional issue types, more detailed rules, notifications, assignment-group routing, reporting, and other ServiceNow automation capabilities.

15. Future Enhancement

- Add more categories and subcategories as new helpdesk issues are identified.
- Introduce automatic Assignment Group routing after classification.
- Add notifications for selected incident types.
- Create reports to measure classification volume and rule effectiveness.
- Maintain a centralized classification-rule table for easier administration.

16. Design Conclusion

The Solution 2 design establishes a structured architecture for Auto Ticket Classification using Flow Designer. It separates incident input, rule evaluation, classification mapping, and record update into clear stages. This approach supports the project objective of automating common IT helpdesk ticket classification while keeping the solution understandable, testable, and scalable.

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